Practical 9 — In-Memory Caching and Query Optimization

❖ What we will build

GET /tasks

| v

Check node-cache

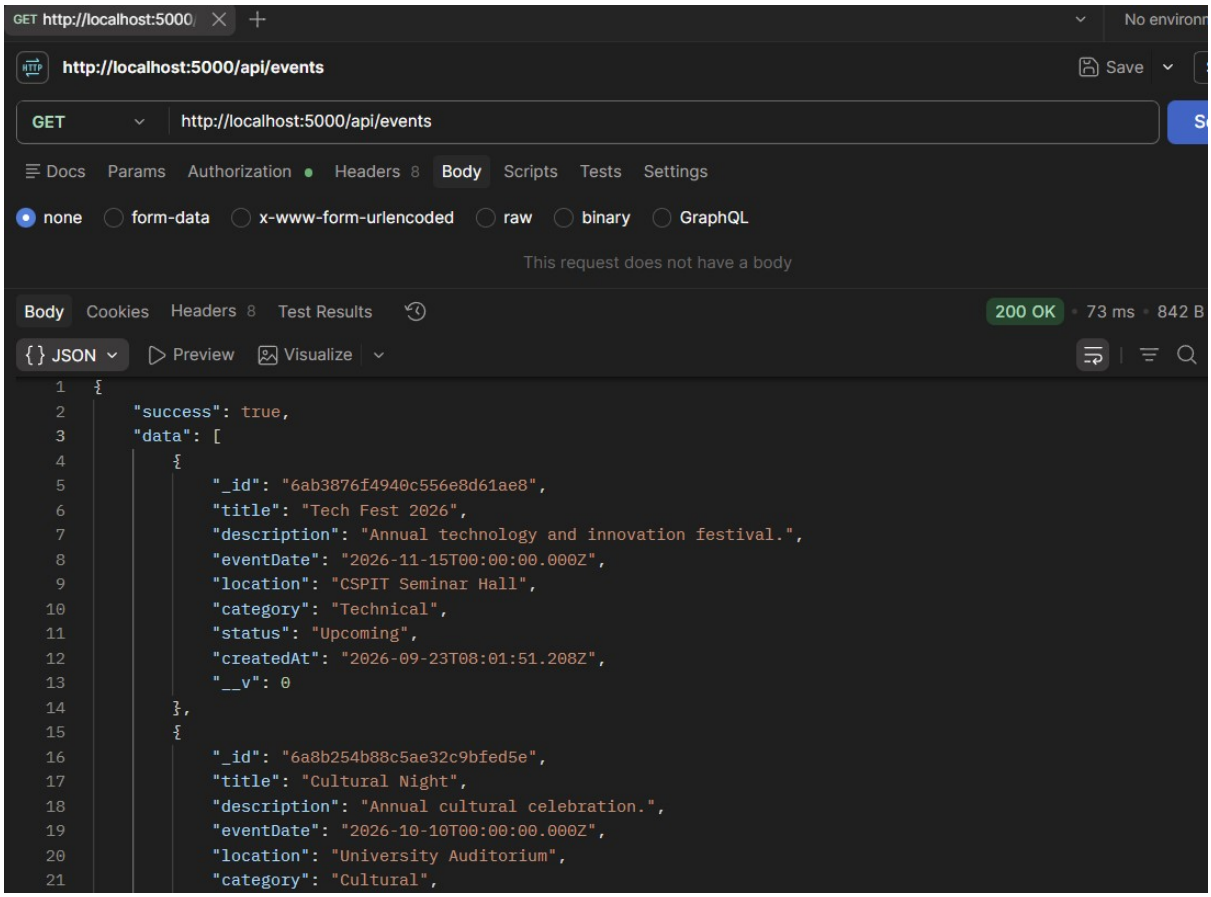
|

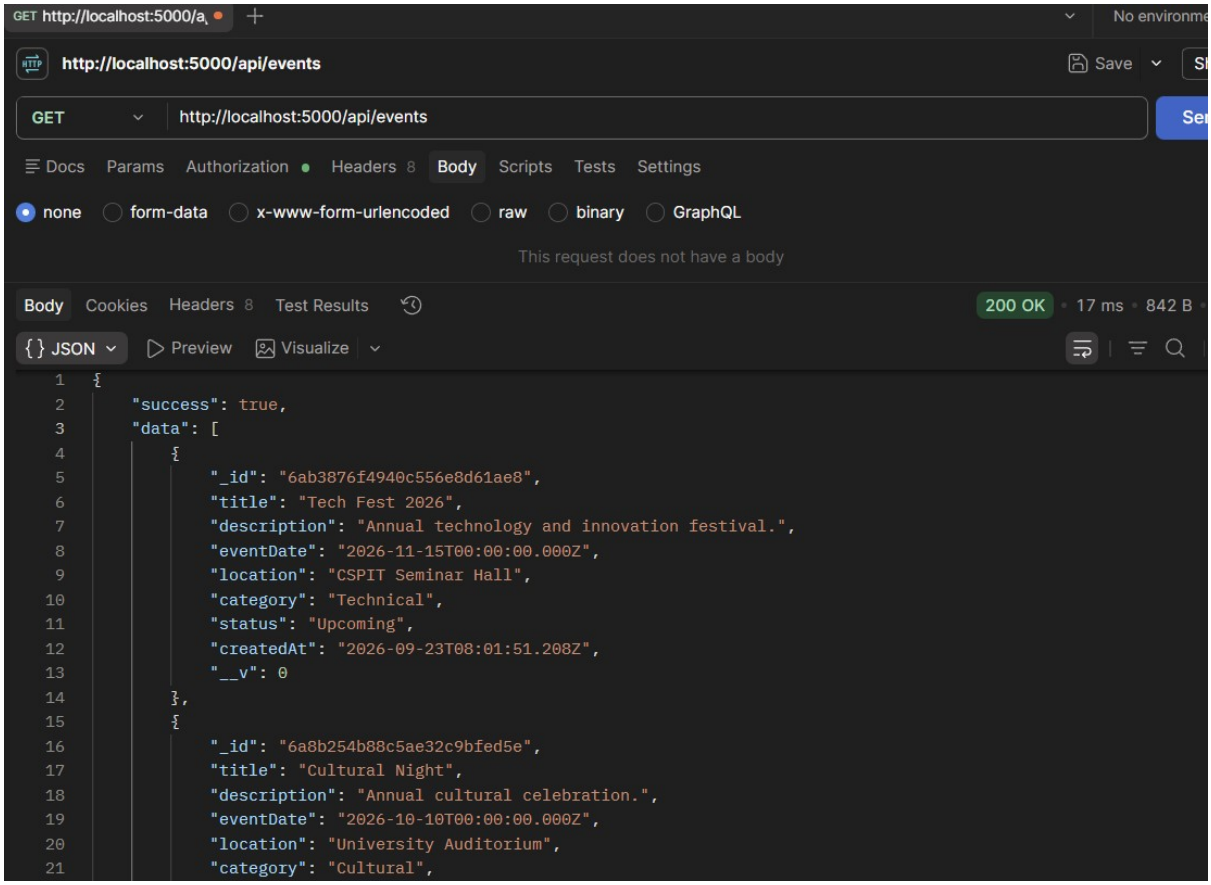
+---- HIT ----> Return cached data

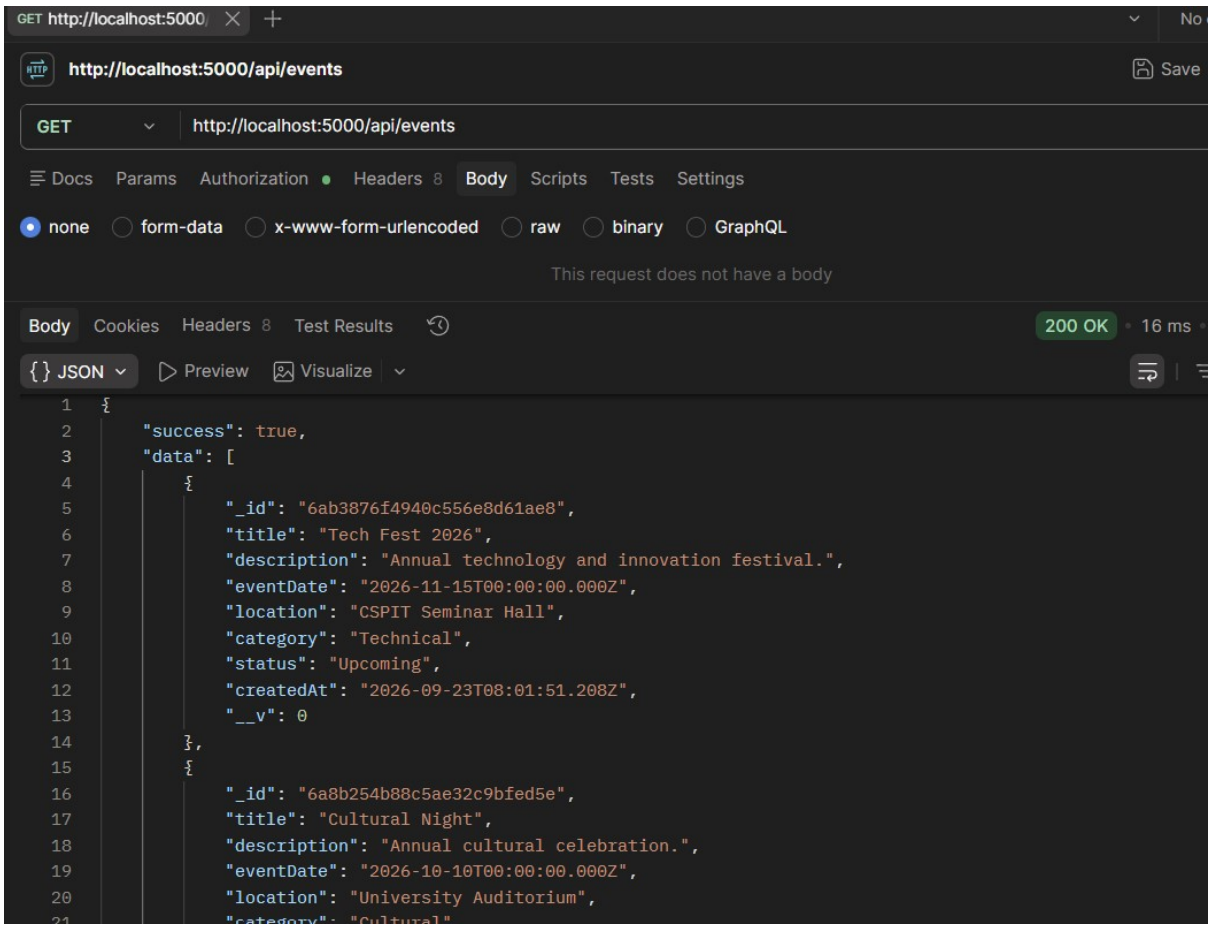
|

+---- MISS ---> MongoDB ---> Store in cache ---> Return data

Measure UNCACHED response time




GET http://localhost:5000/a
http://localhost:5000/api/events
Save
GET
http://localhost:5000/api/events
Docs Params Authorization Headers 8 Body Scripts Tests Settings
none form-data x-www-form-urlencoded raw binary GraphQL
This request does not have a body
Body Cookies Headers 8 Test Results
200 OK 17 ms 842 B
{} JSON Preview Visualize
{
"success": true,
"data": [
{
"_id": "6ab3876f4940c556e8d61ae8",
"title": "Tech Fest 2026",
"description": "Annual technology and innovation festival.",
"eventDate": "2026-11-15T00:00:00.000Z",
"location": "CSPIT Seminar Hall",
"category": "Technical",
"status": "Upcoming",
"createdAt": "2026-09-23T08:01:51.208Z",
"__v": 0
},
{
"_id": "6a8b254b88c5ae32c9bfed5e",
"title": "Cultural Night",
"description": "Annual cultural celebration.",
"eventDate": "2026-10-10T00:00:00.000Z",
"location": "University Auditorium",
"category": "Cultural",


GET http://localhost:5000/
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"createdAt": "2026-09-23T08:01:51.208Z",
"__v": 0
},
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"_id": "6a8b254b88c5ae32c9bfed5e",
"title": "Cultural Night",
"description": "Annual cultural celebration.",
"eventDate": "2026-10-10T00:00:00.000Z",
"location": "University Auditorium",

Lab journal table

Request	Uncached Time
1	73 ms
2	17 ms
3	16 ms

Install node-cache

```
E:\CHARUSAT\SEM 5\AWDF\PRA_6\backend>npm install node-cache

up to date, audited 109 packages in 4s

18 packages are looking for funding
  run `npm fund` for details
```

```
E:\CHARUSAT\SEM 5\AWDF\PRA_6\backend>npm list node-cache
full-stack-management-backend@1.0.0 E:\CHARUSAT\SEM 5\AWDF\PRA_6\backend
└── node-cache@5.1.2


E:\CHARUSAT\SEM 5\AWDF\PRA_6\backend>
```

Create cache.js

JS cache.js U

backend > JS cache.js > ...

```
const NodeCache = require("node-cache");

const cache = new NodeCache({
  stdTTL: 60
});

let cacheHits = 0;
let cacheMisses = 0;

cache.incrementHit = () => {
  cacheHits++;
};

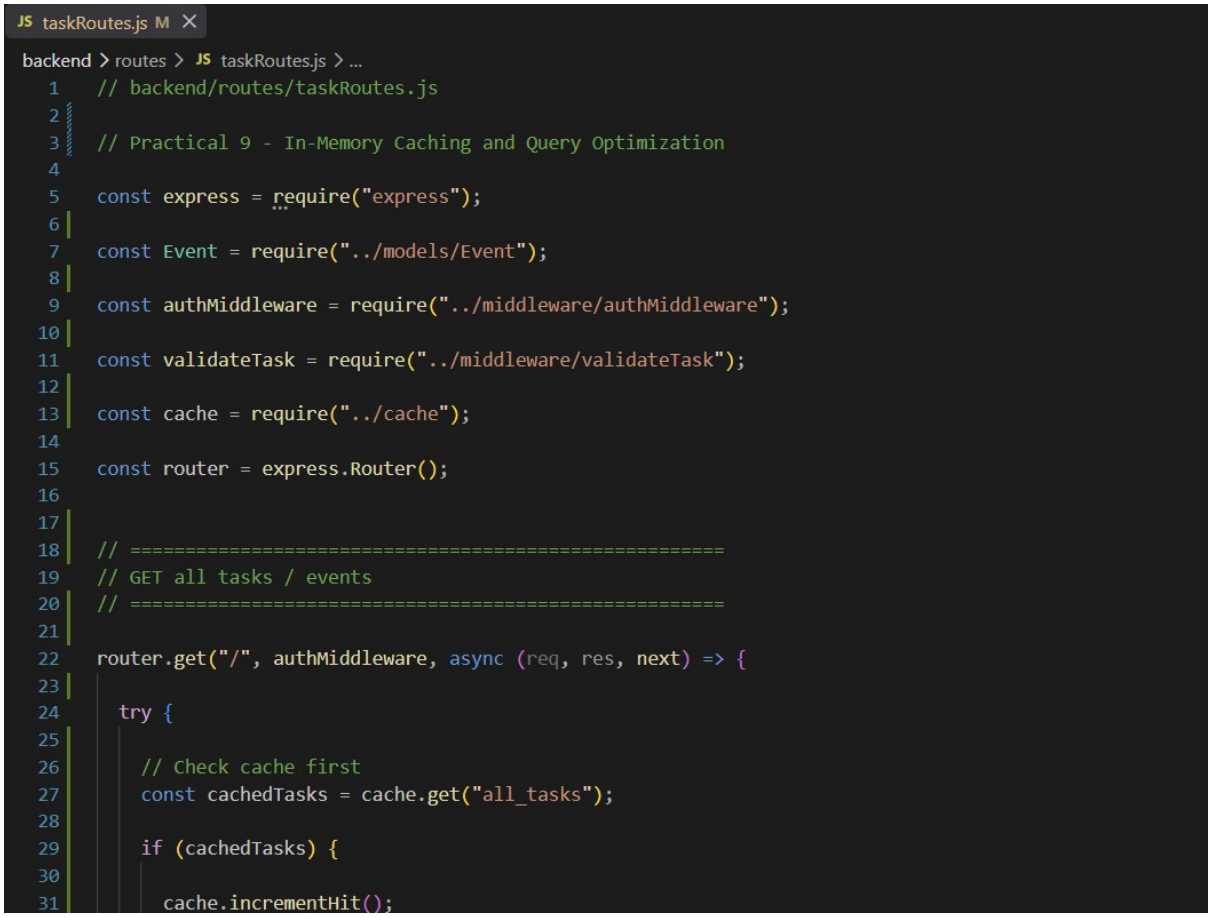
cache.incrementMiss = () => {
  cacheMisses++;
};

cache.getStats = () => {
  return {
    hits: cacheHits,
    misses: cacheMisses
  };
};

module.exports = cache;
```

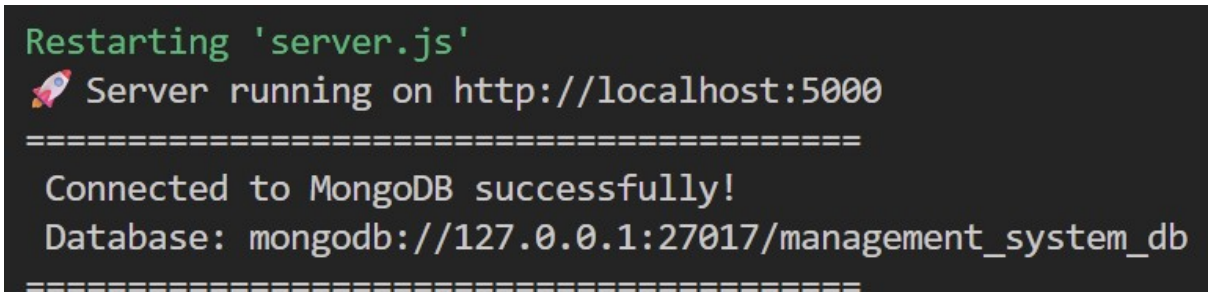
---

update taskRoutes.js

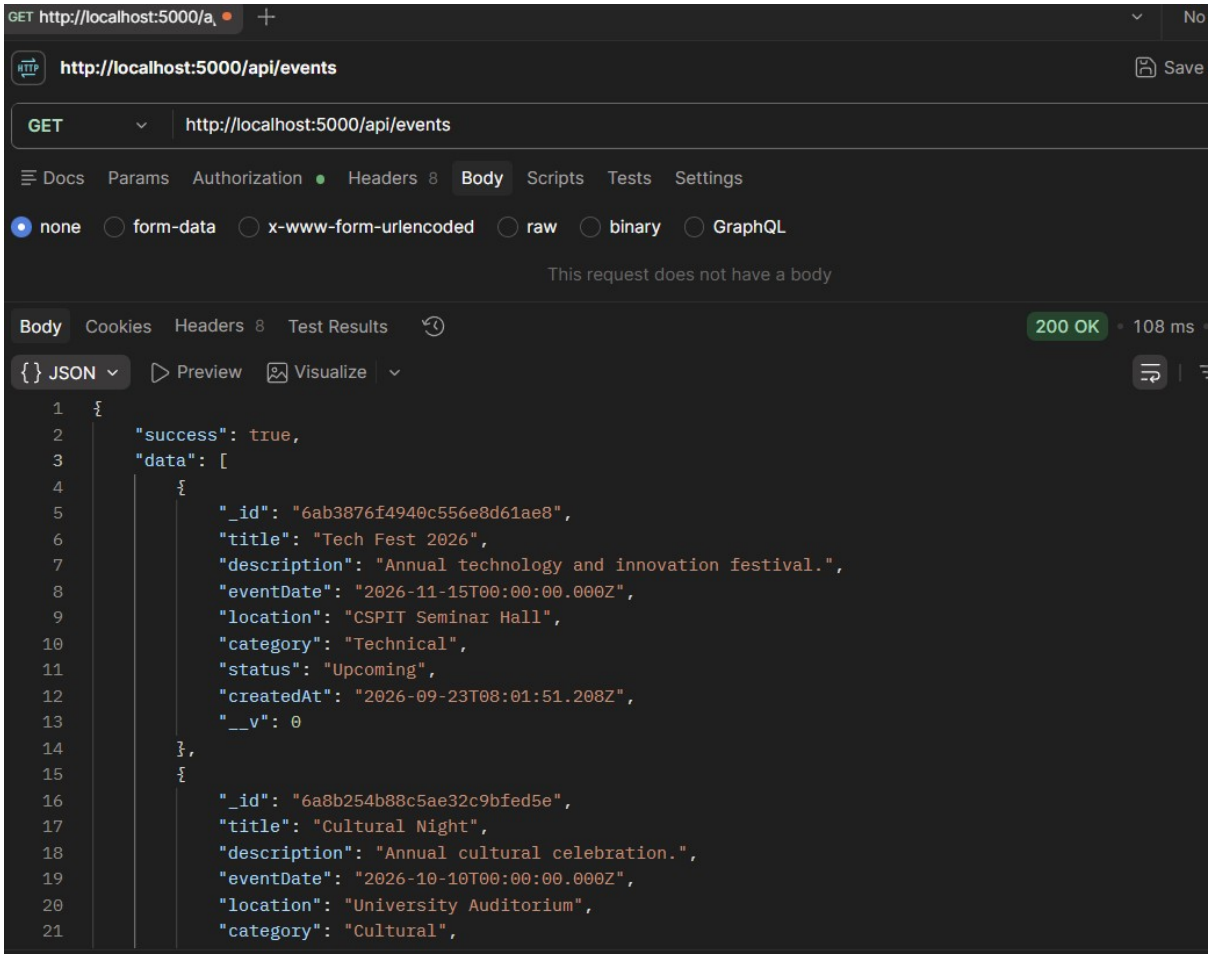


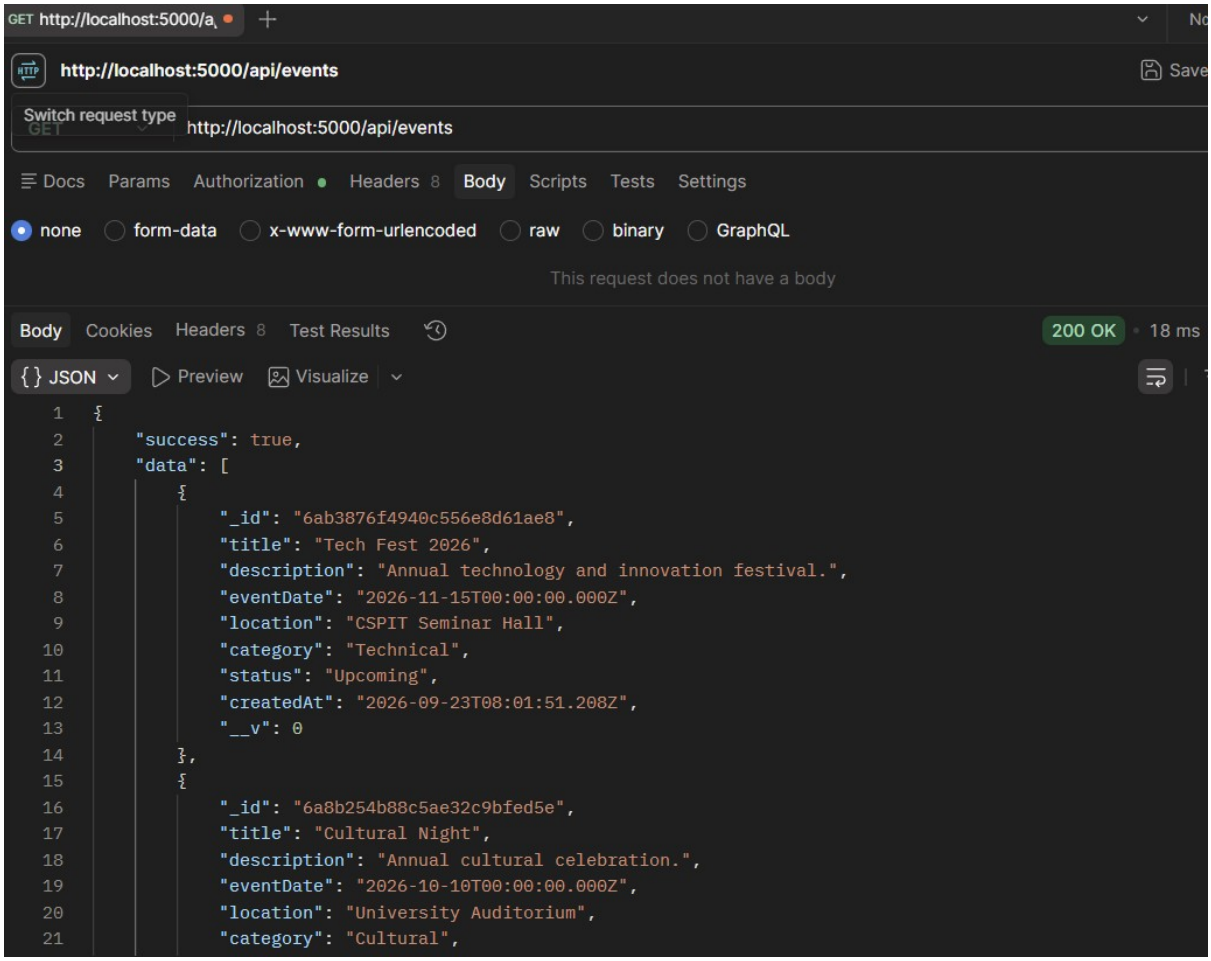
```
JS taskRoutes.js M
backend > routes > JS taskRoutes.js > ...
1  // backend/routes/taskRoutes.js
2
3  // Practical 9 - In-Memory Caching and Query Optimization
4
5  const express = require("express");
6
7  const Event = require("../models/Event");
8
9  const authMiddleware = require("../middleware/authMiddleware");
10
11 const validateTask = require("../middleware/validateTask");
12
13 const cache = require("../cache");
14
15 const router = express.Router();
16
17
18 // ====================================================
19 // GET all tasks / events
20 // ====================================================
21
22 router.get("/", authMiddleware, async (req, res, next) => {
23
24   try {
25
26     // Check cache first
27     const cachedTasks = cache.get("all_tasks");
28
29     if (cachedTasks) {
30
31       cache.incrementHit();
```

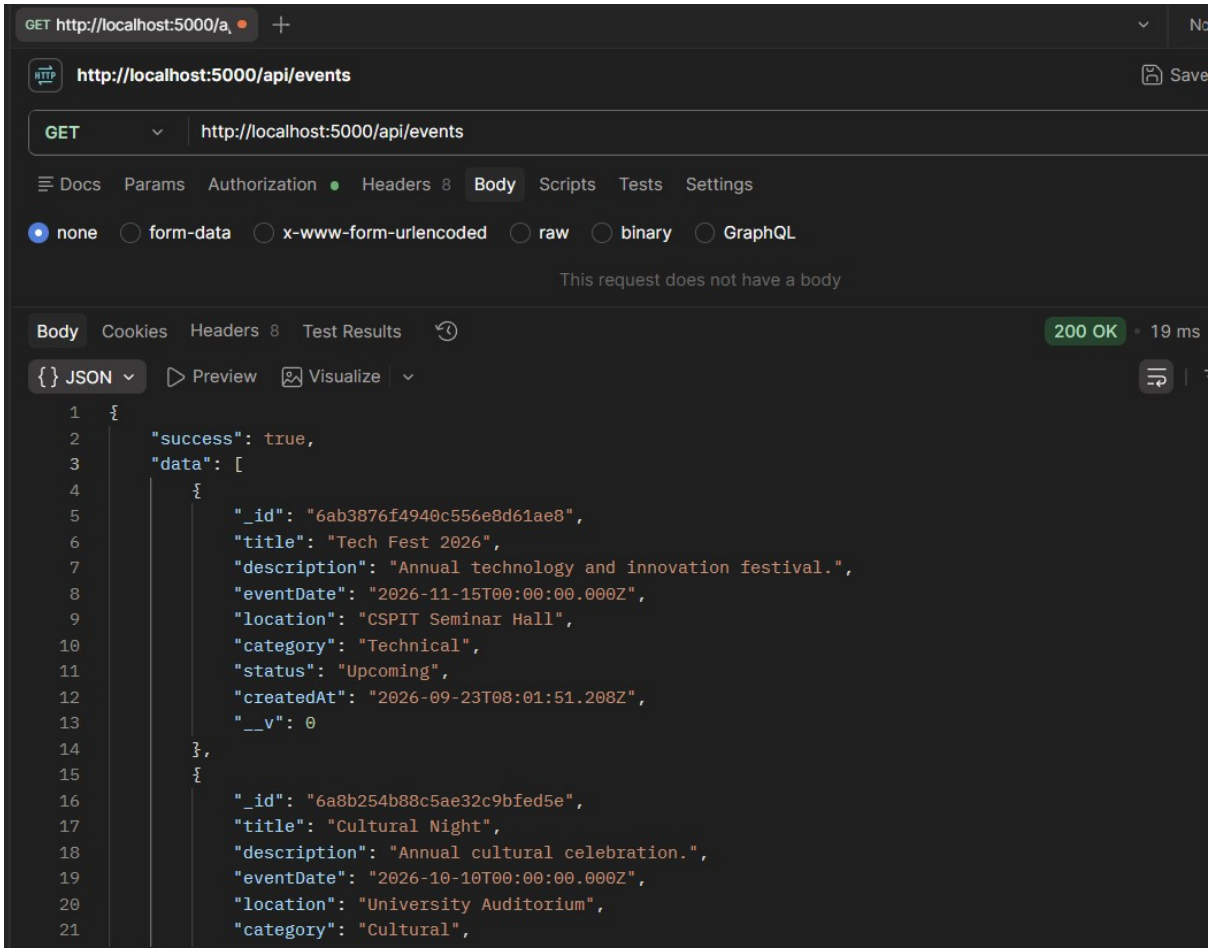
Test the cache



```
Restarting 'server.js'
🚀 Server running on http://localhost:5000
==========================================
 Connected to MongoDB successfully!
 Database: mongodb://127.0.0.1:27017/management_system_db
==========================================
```


GET http://localhost:5000/a
http://localhost:5000/api/events
Save
GET
http://localhost:5000/api/events
Docs Params Authorization Headers 8 Body Scripts Tests Settings
none form-data x-www-form-urlencoded raw binary GraphQL
This request does not have a body
Body Cookies Headers 8 Test Results
200 OK 108 ms
{ } JSON Preview Visualize
{
"success": true,
"data": [
{
"_id": "6ab3876f4940c556e8d61ae8",
"title": "Tech Fest 2026",
"description": "Annual technology and innovation festival.",
"eventDate": "2026-11-15T00:00:00.000Z",
"location": "CSPIT Seminar Hall",
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"__v": 0
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"_id": "6a8b254b88c5ae32c9bfed5e",
"title": "Cultural Night",
"description": "Annual cultural celebration.",
"eventDate": "2026-10-10T00:00:00.000Z",
"location": "University Auditorium",
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GET http://localhost:5000/a
http://localhost:5000/api/events
Save
Switch request type
GET
http://localhost:5000/api/events
Docs Params Authorization Headers 8 Body Scripts Tests Settings
none form-data x-www-form-urlencoded raw binary GraphQL
This request does not have a body
Body Cookies Headers 8 Test Results
200 OK 18 ms
{} JSON Preview Visualize
{
"success": true,
"data": [
{
"_id": "6ab3876f4940c556e8d61ae8",
"title": "Tech Fest 2026",
"description": "Annual technology and innovation festival.",
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First request

Terminal:

Measure CACHED response time

Request	Uncached	Cached
1	73 ms	8 ms
2	17 ms	4 ms
3	16 ms	3 ms

Github: https://github.com/sahilmak1602/WDF_24IT045_SAHIL.git